

Helium Atom

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$$\hat{H} = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{e^2}{4\pi\epsilon_0} \left(\frac{2}{r_1} + \frac{2}{r_2} - \frac{1}{|r_1 - r_2|} \right) \quad (1)$$

If we didn't have the last piece it would be like the Hydrogen atom case. Using Separation of variables. As we are doing variational calculations we make a guess.

$$\psi_{guess}(r_1, r_2) = \phi_0(r_1) \phi_0(r_2) \quad (2)$$

with the single electron wave function (from solving hydrogen Schrodinger equation)

$$\psi_0(r_1) = \frac{4}{\sqrt{2\pi}a_0^{\frac{3}{2}}} e^{-\frac{2r}{a_0}} \quad (3)$$

with the Bohr radius being

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \quad (4)$$

We get a factor of 2 from 2 electrons, another factor of 2 from the fact they are comperssed and a factor of 2 from the fact XXX.

Thus we have $8E_0$ with E_0 the ground state energy of hydrogen and neglecting the $\frac{1}{|r_1 - r_2|}$.
 $-108.8 \text{ eV} = 9E_0$. Experimentally -78.8 eV .

$$E_{\text{trial}} = \langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle = 8E_0 + \frac{e^2}{4\pi\epsilon_0} \langle \psi | \frac{1}{|r_1 - r_2|} | \psi \rangle \quad (5)$$

We call the last term $\langle V_{ee} \rangle$

$$\langle V_{ee} \rangle = \int \int d^3r_1 d^3r_2 \frac{|\psi_0(r_1) \psi_0(r_2)|^2}{|r_1 - r_2|} \quad (6)$$

$$= -\frac{4E_0}{\pi^2} \int \frac{e^{-2(r_1+r_2)}}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}} d^3r_1 d^3r_2 \quad (7)$$

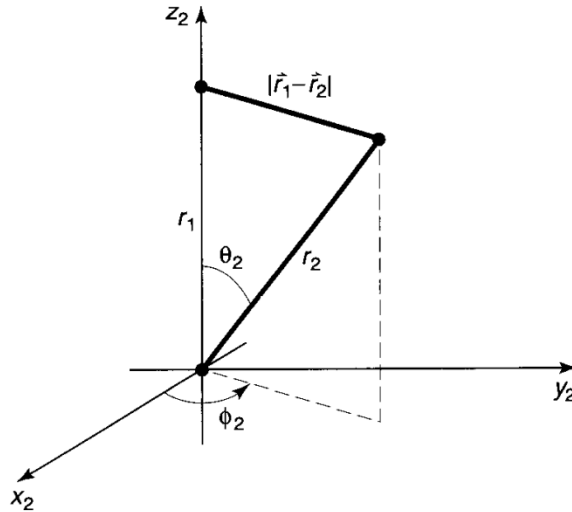


Figure 7.4: Choice of coordinates for the $\mathbf{r}_2$ integral (Equation 7.20).

Figure 1:

$$|r_1 - r_2|^2 = (r_1 - r_2) \cdot (r_1 - r_2) = r_1 \cdot r_1 - 2(r_1 \cdot r_2) + r_2 \cdot r_2 = r_1^2 - 2r_1r_2 \cos \theta + r_2^2 \quad (8)$$

$$= -\frac{4E_0}{\pi^2} \int d^3r_1 e^{-2r_2} \left[\int d^3r_2 \frac{e^{-2r_1}}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}} \right] \quad (9)$$

$$I(r_1) = \left[\int d^3r_2 \frac{e^{-2r_1}}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}} \right] \quad (10)$$

$$I(r_1) = \int r_2^2 \sin \theta dr_2 d\phi_2 d\theta \frac{e^{-2r_2}}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos \theta}} \quad (11)$$

Let $\mu = \cos \theta$ thus $d\mu = \sin \theta d\theta$.

$$I(r_1) = 2\pi \int_0^\infty dr \int_{-1}^1 d\mu \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1r_2\mu}} \quad (12)$$

$$I(r_1) = 2\pi \int_0^\infty dr \left[\frac{\sqrt{r_1^2 + r_2^2 - 2r_1r_2\mu}}{r_1r_2} \right]_1^{-1} \quad (13)$$

$$\left[\frac{\sqrt{r_1^2 + r_2^2 - 2r_1r_2\mu}}{r_1r_2} \right]_1^{-1} = \begin{cases} \frac{2}{r_1} & \text{if } r_1 > r_2 \\ \frac{2}{r_2} & \text{if } r_2 > r_1 \end{cases} \quad (15)$$

$$I(r_1) = 4\pi \left(\frac{1}{r_2} \int_0^{r_2} e^{-2r_1} r_1^2 dr_1 - \int_{r_2}^\infty e^{-2r_1} r_1 dr_1 \right) = I(r) = \frac{\pi}{r} [1 - e^{-2r} (1 + r)] \quad (16)$$

$$< V_{ee} > = -\frac{16\epsilon_0}{\pi} \int_0^\infty e^{-2r} I(r) r^2 dr = -\frac{5}{2} E_0 \quad (17)$$

$$< V_{ee} > = 8E_0 - \frac{5}{2} E_0 = \frac{11}{2} E_0 = -74.8eV \quad (18)$$

Which almost matched the experimental $-78.89eV$. And is an upper-bound